A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Topic-by-topic competency mapping, assessment design and usage guide for AyUG-RB — Second Professional BAMS
Ayurvediya Aushadhi Nirmana & Prayoga Vigyana

Curriculum	NCISM, Board of Ayurveda — applicable from the 2021–22 batch, academic year 2023–24 onwards
Coverage	2 papers · 26 theory topics · 10 practical groups · 268 mapped learning objectives
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3 Column E3 and Table 4 Column E4

450	240	210	400
TEACHING HOURS	THEORY HOURS	PRACTICAL HOURS	TOTAL MARKS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

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Unit 2 Paribhasha (Terminology) 19 obj · Shows how 13

Unit 3 Adharbhuta Siddhanta (Application of fundamental 11 obj · Knows how 15

Unit 4 Yantropakaranani- I (Equipments and machineries) 6 obj · Knows how 17

Unit 5 Yantropakaranani -II (Equipments, fuel and Heating 21 obj · Does 19

Unit 6 Kalpana Nirmana I (Primary & Secondary dosage 11 obj · Knows how 21

Unit 7 Kalpana Nirmana-II (Method of Preparation of 4 obj · Knows how 23

Unit 8 Rasa Dravya Parichaya- I 9 obj · Shows how 25

Unit 9 Rasa Dravya Parichaya II 7 obj · Knows how 27

Unit 10 Rasadravya Parichaya III 5 obj · Knows 29

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Unit 13 Current and emerging trend in Ayurvedic 6 obj · Knows how 35

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PART B — PAPER 2, AYURVEDIYA AUSHADHI PRAYOGA VIGYANA

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Unit 2 Single drug (Herbal & Mineral) 5 obj · Knows how 40

Unit 3 Single drug(Bhasma, Shuddha & Pishti) 5 obj · Shows how 42

Unit 4 Aushadhi Kalpa -I (Compound formulations) 5 obj · Knows how 44

Unit 5 Aushadhi Kalpa-II (Compound Drugs/Formulations) • Chemical/phytochemical composition 7 obj · Knows how 46

Unit 6 Dosage Forms & Cosmetic Products 12 obj · Knows how 48

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PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the units.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column E3
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a unit in Part B**

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PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

This curriculum is weighted towards the lower tiers — 127 of its 268 objectives sit at *Knows* and a further 107 at *Knows how*. Only 34 reach *Shows how* or *Does*. For a second-year undergraduate that is a defensible shape, but it means the practical hours carry almost all of the evidence that cannot be gathered on paper, and they are easy to let slip.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN E3

Open Table 3 (theory) and Table 4 (practical). The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 268 objectives of this curriculum the distribution is:

TIER	OBJECTIVES	SHARE
Knows	127	47.4%
Knows how	107	39.9%
Shows how	23	8.6%
Does	11	4.1%

Total 268 objectives.

Two thirds of this curriculum can be certified on paper. The remaining third cannot, and it is concentrated in the practical groups. If the practical record is treated as an administrative task, that third goes uncertified while the paperwork looks complete.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every unit in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every unit.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a unit to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum names its instruments in Column G3 and G4 — TT-Theory, T-EW, T-OBT, P-VIVA, P-ID, OSPE, checklists, the log book and the practical record. Appendix C3 sorts every one of them by the tier it can honestly reach.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each unit what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a unit in Part B

Part B has twenty-six theory units and ten practical units. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this unit reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this unit** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the unit says where.

Auditing what you already do. Take your existing assessment for a unit and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The units

Thirty-six units: twenty-six following the theory topics of Papers 1 and 2, and ten following the practical topics of Table 4. Each carries the curriculum's own objectives gathered by tier, then the pyramid for that unit — what each tier asks for here, how to build it, how to assess it, and what evidence you keep. Chapter 5 explains the layout.

1. **Paper 1** — Ayurvediya Aushadhi Nirmana Vigyana, units 1 to 14
2. **Paper 2** — Ayurvediya Aushadhi Prayoga Vigyana, units 1 to 12
3. **Practical** — NCISM Table 4, units 1 to 10

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PAPER 1 · UNIT 1

Chronological development of Ayurvediya Aushadhi Nirmana

Hours: 2 lecture · 1 non-lecture | Term 1 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra.	Must know	TT-Theory
Describe about contribution of Nagarjuna Acharya to Rasashastra.	Must know	TT-Theory
Enlist important classical texts of Rasashastra and describe their unique features in short.	Desirable	TT-Theory
Describe Recent development in Ayurvedic Pharmaceuticals viz new dosage forms, pharmaceutical modification techniques.	Desirable	TT-Theory
Define Rasa and Rasayana and describe difference between Rasa and Rasayana.	Desirable	—
KNOWS HOW — 2 OBJECTIVES		
Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus.	Must know	—
Justify design of ancient Rasashala.	Nice to know	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus; Justify design of ancient Rasashala.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra; Describe about contribution of Nagarjuna Acharya to Rasashastra; Enlist important classical texts of Rasashastra and describe their unique features in short; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 2

Paribhasha (Terminology)

Hours: 8 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **19 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
10	8	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 10 OBJECTIVES		
Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana.	Must know	T-CRQs
Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations.	Must know	—
Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya.	Must know	P-ID, P-REC, PUZ
Enlist and discuss Sandigdha, Krutrima, Pratinidhi and Anupalabdha dravya.	Desirable	P-ID
Explain the term Marana and describe its types with examples.	Must know	T-OBT
Enlist all relevant prakriya paribhasha of Amrutikarana Lohitikarana, Sattvapatana, druti and discuss with examples.	Desirable	PRN
Identify names of Parada Bandhas.	Nice to know	T-OBT
Recognise and discuss Dhanvantari Bhaga and Rudra bhaga.	Nice to know	C-INT
Explain importance of ethical practices for drug processing(Shodhan, Marana).	Must know	P-POS
Explain value of keen and accurate application of weights and measures in Ayurvediya Aushadhi nirmana.	Desirable	PRN
KNOWS HOW — 8 OBJECTIVES		
Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana.	Must know	QZ
Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana.	Must know	M-CHT, PRN, T-EMI
Illustrate all Bhasma pariksha as per classical description.	Must know	P-EXAM
Define the word Siddhilakshani. Recite examples of classical siddhilakshani. Interpret its rationality.	Must know	—
Explain importance of Mana-paribhasha and classical types of Mana. Recall Charts of Mana.	Desirable	T-CRQs
Categorize parameters of Drug Standardization and develop a checklist for assessment of quality of rasadravyas.	Desirable	WP
Explain Value of selection of genuine raw material.	Must know	DEB
Discuss and justify importance of Bhasma Pariksha.	Must know	—
SHOWS HOW — 1 OBJECTIVE		
Describe Grahyagrahya parameters used for selection of Rasadravyas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Describe Grahyagrahya parameters used for selection of Rasadravyas.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
8 objectives

Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana; Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana; Illustrate all Bhasma pariksha as per classical description; and 5 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
10 objectives

Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana; Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations; Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya; and 7 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 3

Adharbhuta Siddhanta (Application of fundamental principles)

Hours: 4 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **11 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	9	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms.	Desirable	—
Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono- Therapeutics.	Must know	TT-Theory
KNOWS HOW — 9 OBJECTIVES		
Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas.	Must know	M-POSF
Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected.	Must know	—
Describe time of Dravya collection and explain rationality behind.	Must know	—
Enlist parts of Sthavara & Jangama Dravyas used for Ayurvediya Aushadhi nirmana.	Must know	—
Define Samskara. Discuss role of Samskara in Ayurvediya Aushadhi Nirmana alongwith various examples mhita.	Must know	—
Describe Ardra - Shushka Dravya & Anukta – Visheshokta Dravya collection principles.	Must know	—
Describe importance of kalpa sevan matra. Explain factors considered for deciding dosage of any drug (Ayurvedic as well as modern medicine principles). Describe Posology.	Must know	—
Discuss Yogika Dravya Sidhdhanta(Drug combination).	Must know	—
Justify Aushadhi kalpa namakarana siddhanta with examples.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas; Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected; Describe time of Dravya collection and explain rationality behind; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms; Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono-Therapeutics.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 4

Yantropakaranani- I (Equipments and machineries)

Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Choose and record contemporary machines used in Ayurvedic drug preparation.	Desirable	—
Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana.	Desirable	—
Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra.	Desirable	—
Enlist Ayurvediya aushadhi kalpana and equipments/yantras/ machines used for preparation of each kalpana.	Desirable	—
KNOWS HOW — 2 OBJECTIVES		
Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana.	Must know	—
Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.	Desirable	M-POS

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana; Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Choose and record contemporary machines used in Ayurvedic drug preparation; Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana; Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 5

Yantropakaranani -II (Equipments, fuel and Heating Devices)

Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **21 objectives** | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
13	7	—	1
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 13 OBJECTIVES		
Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana.	Desirable	TT-Theory
Define the term Puta and recite its classical explanation.	Must know	M-POSF
Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa.	Must know	M-POS
Identify and enlist contemporary devices used in the preparation of Bhasma. T.	Desirable	M-POS, QZ
Review the temperature pattern of various Putas by referring research articles.	Desirable	M-POS, QZ
Compile names and significance of temperature measuring devices with reference to Puta.	Desirable	M-POSF
Enlist various devices used for heating during Ayurvediya Aushadhi nirmana.	Desirable	P-SUR
Enlist constituents needed for Samanya Musha Nirmana.	Desirable	—
Enlist and discuss the material used for Sandhi Bandhan (while using Sharava and Kachakupi).	Nice to know	T-OBT
Identify and record the types of Crucibles.	Nice to know	M-POS
Describe types of Koshthi and their uses.	Desirable	COM
Describe the Current trends in heat transfer device e.g. steam jacketed heating device, programmed muffle furnace programmed furnace for Parpati / Pottali preparation.	Desirable	COM, M-POS
Enlist various fuels used for heating for estimation of their heat value.	Nice to know	M-POS
KNOWS HOW — 7 OBJECTIVES		
Recognise the principles behind construction and working of the classical Putas.	Must know	M-POSF
Enlist the specific Puta used for a particular Bhasma Nirmana.	Desirable	M-POSF
Interpret the effect of heat transformation in the material subjected to heating through Musha.	Desirable	PRN
Recall Pakaj Utpatti Siddhant and interpret it for Agni Sannikarsha Sanskara.	Desirable	—
Explain the Pharmaceutical use of the Hot plate, heating mantle induction stove, hot air oven, muffle furnace (horizontal and vertical type).	Nice to know	M-POSF, P-SUR
Assess and interpret the effect of fuel in quantum of heat given (time and temperature).	Desirable	M-POS
Analyse and appraise use of proper heating device and fuels used for Ayurvediya Aushadhi Nirmana.	Desirable	DEB
DOES — 1 OBJECTIVE		
Define the term Musha and describe uses of various types of Musha.	Desirable	T-OBT

THE PYRAMID FOR THIS UNIT

Does 1 objective	Define the term Musha and describe uses of various types of Musha. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how not used here	This unit does not ask for Shows how. Nothing here needs to be certified at that standard. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 7 objectives	Recognise the principles behind construction and working of the classical Putas; Enlist the specific Puta used for a particular Bhasma Nirmana; Interpret the effect of heat transformation in the material subjected to heating through Musha; and 4 more. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 13 objectives	Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana; Define the term Puta and recite its classical explanation; Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa; and 10 more. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 6

Kalpana Nirmana I (Primary & Secondary dosage forms)

Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **11 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana.	Must know	—
Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples.	Must know	—
Assess and discuss modern day development and market trend of the dosage form.	Desirable	—
Determine the advantages and disadvantages of the dosage form.	Desirable	—
Describe method of preparation of dosage form along with the principle involved.	Must know	—
Describe meaning of the term Upakalpana. Enlist panchavidha kashaya kalpana and their respective Upakalpana.	Desirable	—
Explain various kalpas belonging to Various dosage forms -Ingredients, proportion, matra, sevan vidhi if any, therapeutic.	Must know	P-EXAM
KNOWS HOW — 4 OBJECTIVES		
1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana.	Must know	—
Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc.	Must know	—
Explain in details about Standard Operating Procedure (SOP) of each kalpana.	Must know	—
Critically evaluate rationality behind different proportion of water used in various formulations.	Desirable	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana; Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc; Explain in details about Standard Operating Procedure (SOP) of each kalpana; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana; Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples; Assess and discuss modern day development and market trend of the dosage form; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 7

Kalpana Nirmana-II (Method of Preparation of different dosage forms& Dietary Supplements))

Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates.	Must know	T-OBT
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey.	Must know	—
Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates.	Must know	T-OBT
KNOWS HOW — 1 OBJECTIVE		
Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals.	Must know	P-EXAM

THE PYRAMID FOR THIS UNIT

Does not used here	This unit does not ask for Does. Nothing here needs to be certified at that standard. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how not used here	This unit does not ask for Shows how. Nothing here needs to be certified at that standard. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 1 objective	Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 3 objectives	Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates; Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey; Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 8

Rasa Dravya Parichaya- I

Hours: 12 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **9 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	5	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Enlist synonyms of Rasadravya and explain significance of that T.	Must know	—
Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya.	Desirable	PRN
Develop checklist for identification of genuine rasadravyas.	Must know	P-PS
KNOWS HOW — 5 OBJECTIVES		
Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya.	Must know	P-VIVA
Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya.	Must know	P-VIVA
Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha.	Must know	P-EXAM
Explain research updates about Shilajatu, Naga, Mukta and Kapardika.?	Desirable	P-VIVA
Discuss probable physical, chemical changes occurring after Bhavana, Puta & whole process of Bhasma nirmana from Rasadravyas.	Must know	T-OBT
SHOWS HOW — 1 OBJECTIVE		
Explain classical & Mineralogical identification and Sources of Rasadravyas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does not used here	This unit does not ask for Does. Nothing here needs to be certified at that standard. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how 1 objective	Explain classical & Mineralogical identification and Sources of Rasadravyas. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 5 objectives	Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya; Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya; Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha; and 2 more. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 3 objectives	Enlist synonyms of Rasadravya and explain significance of that T; Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya; Develop checklist for identification of genuine rasadravyas. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 9

Rasa Dravya Parichaya II

Hours: 7 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	3	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Explain classical & Mineralogical identification and their Sources.	Desirable	P-VIVA
Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas.	Desirable	P-VIVA
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	Desirable	P-VIVA
Enlist names of rasadravyas and important kalpas prepared from respective dravya. Describe therapeutic importance of that Rasadravya.	Desirable	M-CHTF
KNOWS HOW — 3 OBJECTIVES		
Enlist the important synonyms of Rasadravyas.	Desirable	P-VIVA
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	Must know	—
Explain research updates of Kasisa, Gauripashana and Akika.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Enlist the important synonyms of Rasadravyas; Explain Shodhana, Marana & and other processing techniques with probable chemical reactions; Explain research updates of Kasisa, Gauripashana and Akika.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Explain classical & Mineralogical identification and their Sources; Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas; Explain Shodhana, Marana & and other processing techniques with probable chemical reactions; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 10

Rasadravya Parichaya III

Hours: 3 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **5 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Enlist Synonyms and sources.	Nice to know	M-CHTF, T-OBT
Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas.	Nice to know	CHK, P-VIVA
Explain Shodhana, Marana and Probable Chemical Changes.	Desirable	PRN
Explain research update of Badarashma.	Nice to know	P-VIVA
Enlist Rasadravyas from this group and describe important kalpas with their therapeutic utility.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Enlist Synonyms and sources; Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas; Explain Shodhana, Marana and Probable Chemical Changes; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 11

Kalpana Nirman -III (Method of Preparation of different dosage forms)

Hours: 8 lecture · 6 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life.	Nice to know	P-VIVA
Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life.	Nice to know	P-VIVA
Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques.	Desirable	T-CRQs
Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	Must know	V-KC
KNOWS HOW — 2 OBJECTIVES		
Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques.	Must know	P-VIVA
Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.	Nice to know	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques; Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life; Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life; Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 12

Chaturvidha Rasayana

Hours: 4 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	4	1	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 4 OBJECTIVES		
Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana.	Must know	—
Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana.	Desirable	—
Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana.	Desirable	—
Explain law of Definite proportion and proportion of Mercury and sulphur needed for preparation of sulphide of Mercury. in Chaturvidha rasayana.	Desirable	—
SHOWS HOW — 1 OBJECTIVE		
Explain shelf-life of Chaturvidha Rasayana.	Desirable	P-VIVA
DOES — 1 OBJECTIVE		
Explain definition, types and method of preparation of Chaturvidha Rasayana.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
1 objective

Explain definition, types and method of preparation of Chaturvidha Rasayana.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Explain shelf-life of Chaturvidha Rasayana.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana; Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana; Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 13

Current and emerging trend in Ayurvedic pharmaceuticals

Hours: 3 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Discuss the classification of different dosage forms.	Desirable	M-POSF, P-VIVA
Explain the need of different dosage forms.	Desirable	M-POSF, P-VIVA
Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products.	Desirable	M-POS, PRN
Choose and record the contemporary machineries used in the manufacture of cosmetics.	Nice to know	M-POS
Enlist Quality Control parameters of cosmetics preparations.	Desirable	M-POSF, P-VIVA
KNOWS HOW — 1 OBJECTIVE		
Appreciate Ayurvedic cosmetic products.	Desirable	P-SUR, PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Appreciate Ayurvedic cosmetic products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Discuss the classification of different dosage forms; Explain the need of different dosage forms; Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 14

GMP(Schedule T) & Regulatory aspects of Ayurvedic drugs

Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs.	Must know	—
Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs.	Must know	QZ
Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA.	Desirable	—
Determine the principles and practice of establishment of Ayurvedic pharmacy.	Desirable	—
Discuss the NABL guidelines for testing laboratory (Chemical).	Nice to know	CHK
Explain long forms of these - (FSSAI) and (FDA), (CCRAS), (DCGI), (CDSCO).	Desirable	T-EMI
Discuss ethical aspect of large scale drug preparation in Ayurvedic Pharmacies.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs; Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs; Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 1

Aushadhi Prayoga Vigyana

Hours: 1 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **1 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.	Must know	TT-Theory

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 2

Single drug (Herbal & Mineral)

Hours: 8 lecture · 2 non-lecture | Term 1 | Theory marks: 10 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Appreciate how the therapeutic efficacy varies depending on the dosage form.	Must know	DEB
Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy.	Nice to know	P-VIVA
Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.	Must know	P-VIVA
KNOWS HOW — 2 OBJECTIVES		
Describe different dosage forms prepared out of a single herb or mineral.	Must know	COM, P-VIVA, TT-Theory
Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.	Desirable	COM

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe different dosage forms prepared out of a single herb or mineral; Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Appreciate how the therapeutic efficacy varies depending on the dosage form; Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy; Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 3

Single drug(Bhasma, Shuddha & Pishti)

Hours: 12 lecture · 6 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	3	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.	Desirable	PRN
KNOWS HOW — 3 OBJECTIVES		
Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma.	Desirable	P-VIVA
Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma.	Must know	—
Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.	Desirable	P-VIVA
SHOWS HOW — 1 OBJECTIVE		
Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma; Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma; Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 4

Aushadhi Kalpa -I (Compound formulations)

Hours: 16 lecture · 4 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.	Desirable	P-VIVA
KNOWS HOW — 4 OBJECTIVES		
Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation.	Must know	—
Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation.	Must know	—
Interpret probable mode of action of each formulation as per Ayurveda.	Desirable	—
Critically analyse compound drugs for their therapeutic actions mentioned in the classics.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation; Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation; Interpret probable mode of action of each formulation as per Ayurveda; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 5

Aushadhi Kalpa-II (Compound Drugs/Formulations)

Hours: 14 lecture · 2 non-lecture | Term 3 | Theory marks: 15 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita.	Desirable	P-VIVA
Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given.	Desirable	—
Enlist the formulations with its dose, anupana and indications.	Nice to know	T-EMI
KNOWS HOW — 4 OBJECTIVES		
Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika.	Must know	—
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava.	Nice to know	P-VIVA
Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population.	Must know	P-EXAM, TT-Theory, V-KC
Describe ingredients, with its dose, therapeutic importance anupana and pathya-apathya along with its clinical evidences and Research updates.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika; Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava; Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita; Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given; Enlist the formulations with its dose, anupana and indications.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 6

Dosage Forms & Cosmetic Products

Hours: 5 lecture · 2 non-lecture | Term 3 | Theory marks: 5 | **12 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
11 certified here	1 certified here	— not used	— not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 11 OBJECTIVES		
Define the term- Dosage forms.	Must know	TT-Theory
Discuss the need of different dosage forms.	Must know	TT-Theory
Explain the classification of different dosage forms.	Desirable	TT-Theory
Enlist the solid / liquid / semisolid dosage forms.	Must know	T-EMI
Discuss the routes of administration of different dosage forms.	Must know	—
Describe the advantages and disadvantages of currently available dosage forms.	Must know	TT-Theory
Discuss the research updates about modification of classical Ayurvedic dosage forms.	Desirable	—
Discuss the relevant case studies of different dosage forms.	Nice to know	P-SUR
Define the term Cosmetics.	Desirable	TT-Theory
Explain the classification of cosmetics based on their application on the specific body parts.	Desirable	—
Differentiate between the ancient cosmetics and modern day cosmetics.	Desirable	TT-Theory
KNOWS HOW — 1 OBJECTIVE		
compare and appraise herbal cosmetics vs synthetic products.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

compare and appraise herbal cosmetics vs synthetic products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
11 objectives

Define the term- Dosage forms; Discuss the need of different dosage forms; Explain the classification of different dosage forms; and 8 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 7

Nutraceuticals

Hours: 6 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **9 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Describe importance of combination of aahara and aushadha.	Must know	—
Explain Nutraceuticals with its types.	Must know	TT-Theory
Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage.	Must know	—
Explain research updates about nutraceuticals.	Nice to know	M-POS
Enlist atleast two examples of Aushadhi siddha aahara with its indications for each category-->(Aharadravya belonging to following category and kalpa prepared from that aahara dravya and other aushadhi ingredients)1) Jala varga2) Dugdha varga3) Ikshu varga4) Suka-dhanya varga5) Shimbi Dhanya Varga6) Shaka Varga7) Phala Varga8) Lavana Varga9) Spices mhita.	Desirable	—
KNOWS HOW — 4 OBJECTIVES		
Corelate rasayana and nutraceutical with examples.	Desirable	T-OBT
Describe with examples mode of action of Ayurvedic nutraceuticals mhita.	Must know	PRN
Describe potential of Ayurvedic diet and Rasayana in sports endeavor.	Desirable	DEB
creat list of unique features of Ayurvedic neutraceuticals and appraise their current need.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Corelate rasayana and nutraceutical with examples; Describe with examples mode of action of Ayurvedic nutraceuticals mhita; Describe potential of Ayurvedic diet and Rasayana in sports endeavor; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Describe importance of combination of aahara and aushadha; Explain Nutraceuticals with its types; Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 8

Anupana Prayoga for Aushadhi Kalpa

Hours: 4 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Define and classify Anupana and Sahapana.	Must know	—
Describe value of proper selection of Anupanas with examples.	Desirable	M-CHT
KNOWS HOW — 2 OBJECTIVES		
Explain different factors to be considered for selection of Anupana as per Disease and Patients.	Must know	—
Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Explain different factors to be considered for selection of Anupana as per Disease and Patients; Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Define and classify Anupana and Sahapana; Describe value of proper selection of Anupanas with examples.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 9

Aushadhi Prayoga Marga

Hours: 5 lecture · 1 non-lecture | Term 3 | Theory marks: 10 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Discuss the various types of Aushadhi Prayog Marga (route of drug administration).	Must know	—
Describe the advantages and disadvantages of each Aushadhi Prayog Marga.	Desirable	P-SUR
Enlist the dosage forms used in the specific Aushadhi Prayog Marga.	Desirable	T-EMI
Appraise administration of drugs through various routes mentioned in Ayurvedic treatment.	Must know	—
KNOWS HOW — 1 OBJECTIVE		
Discuss the nature of drug (s) administered in various routes of drug administration.	Nice to know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Discuss the nature of drug (s) administered in various routes of drug administration.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Discuss the various types of Aushadhi Prayog Marga (route of drug administration); Describe the advantages and disadvantages of each Aushadhi Prayog Marga; Enlist the dosage forms used in the specific Aushadhi Prayog Marga; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 10

Rational prescription along with safe dispensing of Ayurvedic formulations.

Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	3	2	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain the importance of rational prescribing of drugs and the concept of essential drugs.	Desirable	CHK, QZ, V-KC
KNOWS HOW — 3 OBJECTIVES		
Explain the safe dispensing and efficacious use of Ayurvedic drugs.	Must know	T-CS, V-KC
Describe the standard protocol for safe dispensing of Ayurvedic drugs.	Desirable	T-OBT
explain ethical aspects related to prescription writing.	Must know	—
SHOWS HOW — 2 OBJECTIVES		
Describe and write demo ideal prescription.	Must know	CHK, V-KC
Demonstrate and educate home remedies to small group of population.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Describe and write demo ideal prescription; Demonstrate and educate home remedies to small group of population.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain the safe dispensing and efficacious use of Ayurvedic drugs; Describe the standard protocol for safe dispensing of Ayurvedic drugs; explain ethical aspects related to prescription writing.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain the importance of rational prescribing of drugs and the concept of essential drugs.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 11

Traditional & Local health Practices

Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Identify Local Health Traditions and Healing Knowledge.	Desirable	P-SUR
Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?	Desirable	—
Appraise traditional knowledge of Ayurvedic medicines.	Desirable	—
KNOWS HOW — 1 OBJECTIVE		
Recognise the possible potential of product development and research based on Traditional knowledge.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Recognise the possible potential of product development and research based on Traditional knowledge.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Identify Local Health Traditions and Healing Knowledge; Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?; Appraise traditional knowledge of Ayurvedic medicines.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 12

Pharmacovigilance for Ayurveda drugs

Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs.	Desirable	C-INT, M-POS, V-KC
Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs.	Must know	V-KC
Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor.	Desirable	QZ, V-KC, VV-Viva
Debate on ADR of ASU drugs.	Desirable	DEB
KNOWS HOW — 2 OBJECTIVES		
Identify and monitor ADRs.	Desirable	PRN, RK, V-KC
Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.	Desirable	QZ, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Identify and monitor ADRs; Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs; Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs; Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 1

Paribhasha concept based Practicals

9 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	5	—	3
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).	Must know	P-VIVA
KNOWS HOW — 5 OBJECTIVES		
Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya.	Must know	P-ID, P-VIVA
Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka.	Desirable	P-VIVA
Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals.	Desirable	P-VIVA
Identify various weights and recite essential measures from conversion chart of AFI. Recognise weighing machines and their weighing capacity.	Desirable	P-SUR, PRN
Identify and value SOP of each procedure of the practical conducted.	Must know	P-EXAM
DOES — 3 OBJECTIVES		
Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations.	Must know	CHK, P-PRF
Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali.	Must know	P-EXAM, P-REC
Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.	Must know	CHK, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
3 objectives

Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations; Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali; Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya; Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka; Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 2

Panchavidha Kashaya Kalpana & their Upakalpana Practicals

4 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	2	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.	Must know	P-PRF
KNOWS HOW — 2 OBJECTIVES		
Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations.	Must know	P-VIVA
Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.	Must know	P-VIVA
SHOWS HOW — 1 OBJECTIVE		
Demonstrate the method of preparation to get the desired dosage form following SOP.	Must know	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Demonstrate the method of preparation to get the desired dosage form following SOP.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations; Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 3

Rasa Dravya Aushadhi Nirmana Practicals

7 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	3	3	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 3 OBJECTIVES		
Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya.	Must know	P-VIVA
Identify, Utilize & know mechanism of Yantra used for drug preparation.	Must know	VV-Viva
Appraise peculiarities of chaturvidha rasayana preparation.	Desirable	M-POS
SHOWS HOW — 3 OBJECTIVES		
Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation.	Must know	P-EXAM
Demonstrate Sidhdhi Lakshna.	Must know	P-VIVA
Assess finished product as per classics.	Must know	P-VIVA
DOES — 1 OBJECTIVE		
Identify raw Dravya, used part with classical Mana and interpret in metric system.	Must know	P-EXAM

THE PYRAMID FOR THIS UNIT

Does
1 objective

Identify raw Dravya, used part with classical Mana and interpret in metric system.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation; Demonstrate Sidhdhi Lakshna; Assess finished product as per classics.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya; Identify, Utilize & know mechanism of Yantra used for drug preparation; Appraise peculiarities of chaturvidha rasayana preparation.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 4

Bheshaja Kalpana Practicals -I

14 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	4	3	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques.	Must know	P-VIVA
Identify the useful part of the raw material and demonstrate Sattva preparation method.	Must know	P-VIVA
Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects.	Must know	P-VIVA
Demonstrate and. Explain preparation methods of Masi kalpana for its internal or external application in clinical practice.	Desirable	P-VIVA
Demonstrate preparation and application aspects of Upanaha kalpana.	Desirable	P-EXAM, V-KC
Demonstrate preparations of Sikta taila along with identify the genuine sample of Sikta. Perform and. Describe Malahara kalpana preparation method with its confirmatory tests of end point.	Desirable	P-VIVA
Demonstrate preparation of different types of Malahara with its applied aspect.	Nice to know	P-SUR, V-KC
KNOWS HOW — 4 OBJECTIVES		
Demonstrate preparation of Vati along with drying techniques.	Must know	P-VIVA
Demonstrate preparation of Varti along with drying and packaging techniques.	Must know	P-VIVA
Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita.	Nice to know	PRN, V-KC
Observe demonstration of formulation of Danta manjana.	Nice to know	PRN
SHOWS HOW — 3 OBJECTIVES		
Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance.	Must know	—
Demonstrate Lavana preparation methods, along with its packaging techniques.	Desirable	P-VIVA
Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	Desirable	P-VIVA, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance; Demonstrate Lavana preparation methods, along with its packaging techniques; Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Demonstrate preparation of Vati along with drying techniques; Demonstrate preparation of Varti along with drying and packaging techniques; Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques; Identify the useful part of the raw material and demonstrate Sattva preparation method; Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 5

Bheshaja Kalpana Practicals -II

11 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	3	3
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 5 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha.	Desirable	OSPE, P-EXAM, P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha.	Must know	P-REC, P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava.	Must know	—
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kutajarista/Ashokarishta/Takrarishta.	Must know	P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlist the results, packing techniques and storage of unique formulations for e.g. Kanji/ Madushukta.	Desirable	COM, P-VIVA
SHOWS HOW — 3 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila.	Must know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara.	Nice to know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.	Nice to know	PRN
DOES — 3 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana.	Must know	OSPE, P-EXAM, P-REC
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha.	Must know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.	Must know	OSPE, P-EXAM, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
3 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 6

Dosage Forms & Self-care Products Practicals

2 objectives | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	2	—	—
not used	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 2 OBJECTIVES		
Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters.	Desirable	CHK, P-VIVA
Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.	Desirable	P-EXAM, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters; Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 7

Field Visit/ Study Tour

7 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	1	—	1
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections.	Desirable	COM, M-POS
Differentiate between the ancient and contemporary methods of drug preparation.	Desirable	PRN
Enlist the documents required in Batch Manufacturing of Ayurvedic medicines.	Nice to know	P-SUR
Identify the role of Quality Control instruments and equipments in ensuring a quality Ayurvedic product.	Desirable	DEB
Appreciate the importance of Good Manufacturing Practices and Good Packaging Practices required in Ayurvedic Drug manufacturing.	Desirable	DEB, PRN
KNOWS HOW — 1 OBJECTIVE		
Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.	Desirable	—
DOES — 1 OBJECTIVE		
Prepare own products as per prevalent FDA guidelines.	Desirable	P-EXAM

THE PYRAMID FOR THIS UNIT

Does
1 objective

Prepare own products as per prevalent FDA guidelines.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections; Differentiate between the ancient and contemporary methods of drug preparation; Enlist the documents required in Batch Manufacturing of Ayurvedic medicines; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 8

Hospital IPD Practical

6 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	1	—
not used	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 5 OBJECTIVES		
Review & Observe the case Sheets.	Nice to know	P-SUR, V-KC
Discuss for formulation prescription manner.	Desirable	T-CS, V-KC
Measure dose of prescribed drug.	Must know	—
Enlist Time, Duration, Pathya, Apathya while prescribing drugs.	Desirable	P-PRF
value details and methodical writing of case records of hospital IPD patients.	Must know	—
SHOWS HOW — 1 OBJECTIVE		
Demonstrate & Explain prescription method.	Desirable	P-PS, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Demonstrate & Explain prescription method.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Review & Observe the case Sheets; Discuss for formulation prescription manner; Measure dose of prescribed drug; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 9

Drug Dispensing Practical

4 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	2	1	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 2 OBJECTIVES		
Assess practice of prescription processing and labelling of the drugs.	Desirable	—
Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.	Desirable	—
SHOWS HOW — 1 OBJECTIVE		
Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.	Desirable	PRN, V-KC
DOES — 1 OBJECTIVE		
Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.	Nice to know	P-PRF, V-KC

THE PYRAMID FOR THIS UNIT

Does
1 objective

Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Assess practice of prescription processing and labelling of the drugs; Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 10

Quality Control Practicals

10 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	3	5	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Identify the geological description of minerals: physical parameters.	Desirable	P-ID
Appraise quality control procedures done for Ayurvedic formulations.	Desirable	—
KNOWS HOW — 3 OBJECTIVES		
Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India.	Must know	P-VIVA
Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure.	Desirable	P-VIVA, PRN
Determine iodine value, acid value and saponification value of oils / ghee.	Must know	P-VIVA
SHOWS HOW — 5 OBJECTIVES		
Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms.	Must know	P-VIVA
Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book.	Must know	P-EXAM
Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability.	Must know	P-EXAM
Estimate specific gravity, alcohol content and total solids of Asava & Arishta.	Must know	P-EXAM
Calculate the dose of various dosage forms and their dispensing methods.	Must know	P-SUR

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms; Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book; Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India; Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure; Determine iodine value, acid value and saponification value of oils / ghee.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Identify the geological description of minerals: physical parameters; Appraise quality control procedures done for Ayurvedic formulations.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **5 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every unit in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every unit
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

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PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	127	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	107	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	23	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	11	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	127	47.4%
Knows how	107	39.9%
Shows how	23	8.6%
Does	11	4.1%

Total 268 objectives.

PART C · APPENDIX C2

Ceiling of every unit

The highest tier each unit reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

NO	TITLE	OBJ	CEILING
PAPER 1 — AYURVEDIYA AUSHADHI NIRMANA VIGYANA			
1	Chronological development of Ayurvediya Aushadhi	7	Knows how
2	Paribhasha (Terminology)	19	Shows how
3	Adharbhuta Siddhanta (Application of fundamental	11	Knows how
4	Yantropakaranani- I (Equipments and machineries)	6	Knows how
5	Yantropakaranani -II (Equipments, fuel and Heating	21	Does
6	Kalpana Nirmana I (Primary & Secondary dosage	11	Knows how
7	Kalpana Nirmana-II (Method of Preparation of	4	Knows how
8	Rasa Dravya Parichaya- I	9	Shows how
9	Rasa Dravya Parichaya II	7	Knows how
10	Rasadravya Parichaya III	5	Knows
11	Kalpana Nirman -III (Method of Preparation of	6	Knows how
12	Chaturvidha Rasayana	6	Does
13	Current and emerging trend in Ayurvedic	6	Knows how
14	GMP(Schedule T) & Regulatory aspects of Ayurvedic	7	Knows
PAPER 2 — AYURVEDIYA AUSHADHI PRAYOGA VIGYANA			
1	Aushadhi Prayoga Vigyana	1	Knows
2	Single drug (Herbal & Mineral)	5	Knows how
3	Single drug(Bhasma, Shuddha & Pishti)	5	Shows how
4	Aushadhi Kalpa -I (Compound formulations)	5	Knows how
5	Aushadhi Kalpa-II (Compound Drugs/Formulations) • Ch	7	Knows how
6	Dosage Forms & Cosmetic Products	12	Knows how
7	Nutraceuticals	9	Knows how
8	Anupana Prayoga for Aushadhi Kalpa	4	Knows how
9	Aushadhi Prayoga Marga	5	Knows how
10	Rational prescription along with safe dispensing of	6	Shows how
11	Traditional & Local health Practices	4	Knows how
12	Pharmacovigilance for Ayurveda drugs	6	Knows how
PRACTICAL UNITS — NCISM TABLE 4			
1	1.Paribhasha concept based Practicals	9	Does
2	2.Panchavidha Kashaya Kalpana & their Upakalpana Pra	4	Shows how
3	3.Rasa Dravya Aushadhi Nirmana Practicals	7	Does
4	4.Bheshaja Kalpana Practicals -I	14	Shows how

NO	TITLE	OBJ	CEILING
5	5.Bheshaja Kalpana Practicals -II	11	Does
6	6.Dosage Forms & Self-care Products Practicals	2	Knows how
7	7.Field Visit/ Study Tour	7	Does
8	8.Hospital IPD Practical	6	Shows how
9	9. Drug Dispensing Practical	4	Does
10	10.Quality Control Practicals	10	Shows how

A written paper can certify Knows and Knows how. Anything above must be certified in the practical, the record book or the log book.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	TT-Theory 16, T-OBT 10, T-EMI 5, T-CRQs 3, P-ID 4, V-KC 19, M-CHT 2	Recall only. Cannot show reasoning or performance.
Knows how	T-EW, T-MEQs, T-CS 2, P-VIVA 57, VV-Viva 2, C-INT 2, M-POS 14, P-POS 1	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE 6, P-PRF 4, CHK 7, P-MOD	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, COM 6, P-REC 4	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Course curriculum for Second Professional BAMS — Rasashastra evam Bhaishajyakalpana (AyUG-RB).* New Delhi: NCISM. Tables 2, 3, 4 and 4a are the source of every objective, tier, priority and instrument in this book. All 268 objectives and the 100 marks of each paper reconcile against the curriculum's own summary.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.